

AI Campus Navigation Agent

1. Problem Statement

The AI Campus Navigation Agent helps university students find routes between campus locations. The student provides a starting location and a destination, and the agent provides clear step-by-step navigation instructions using available campus route information.

Example:

Student Input: Science Block to Library

2. Problem Formulation

- **Initial State**
 - The student's current location where the journey begins.
 - **Example:** Science Block
- **Goal State / Goal Test**
 - The destination the student wants to reach.
 - **Example:** Library
 - The goal test checks whether the current location is equal to the destination.
- **Actions**
 - Moving from one connected campus location to another.
 - **Example:**
Science Block → Library
- **Transition Model**
 - Describes how an action changes the current state.
 - **Example:**
Science Block + Move to Library → Library
- **Path Cost**
 - The total cost of reaching the destination. In this project, the number of moves or steps can be used as the path cost.
- **State Space**
 - All campus locations known to the agent.
 - **Example:**
Main Gate, CS Department, Library, Science Block, Cafeteria, Computer Lab, Auditorium.
- **Search Space**
 - All possible routes between the available campus locations.

- **Example:** CS Department → Library
CS Department → Cafeteria → Library

3. PEAS

Component	Description
Performance Measure	Correct route, clear instructions, and useful response
Environment	University campus
Actuators	Text-based navigation instructions
Sensors	Student's starting location, destination, and stored route information

4. Agent Type

Goal-Based Agent

The agent is goal-based because its main goal is to guide the student from the starting location to the requested destination.

5. Environment Type

- **Partially Observable** – The agent does not have complete real-time information about the campus.
- **Single-Agent** – One software agent performs the navigation task.
- **Sequential** – Navigation consists of a sequence of steps.
- **Stochastic** – Real campus conditions may change unexpectedly.
- **Dynamic** – Campus conditions can change over time.
- **Discrete** – Locations are represented as separate states.

6. Sensors & Actuators

Sensors

- Student's starting location
- Student's destination
- Stored campus route information

Actuators

- Text-based step-by-step navigation instructions displayed to the student.

7. Rationality

The agent is rational because it selects an available route that matches the student's destination and provides clear navigation instructions based on the available route information. If a route is unavailable, the agent does not invent directions and informs the student that the route is unavailable.